What finite-time computations reveal about graph curvature flows

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Abstract

Certain discrete Ollivier–Ricci flows converge to constant curvature under assumptions; finite runs need their own reading. Four quantities are conflated: finite-horizon residual, numerical counterpart, trend, and the infimum V_G^* (not computed). First, the stopping rule divides the per-step variance change by its initial value, a ratio that shrinks with elapsed time: 27 of the 28 runs it labelled saturated were still falling over the final quarter. Second, a disconnected graph's pooled variance includes a between-component term; within components it falls on 34/34 (median 5.7×). Third, at matched physical time the residual is largely stable to step size for the tested non-converged trajectories; extending three runs from 2,000 to 5,000 steps moves it by <0.2%, and fixed-ε regularisation is not scale invariant. A finite-time residual is a measurement, not a convergence certificate: breaking the curvature–edge map raises it (8/8), and the tested descriptors do not predict it across families.

Introduction

Ricci flow deforms a Riemannian metric in the direction of its Ricci curvature and formed the basis of the uniformisation programme in geometry, culminating in Hamilton's and Perelman's results on three-manifolds [1, 2, 3]. Its discrete counterpart on graphs replaces the metric by an edge-weight function and the curvature by a discrete curvature, most commonly the Ollivier–Ricci curvature κ [4, 5], and updates each edge weight in the direction of its curvature. "Graph curvature" is not a single object: the Bakry–Émery, Ollivier and Forman discretisations give different values on the same graph [6, 7, 8, 9, 10], and further discretisations exist that are not equivalent to any of them [11, 12], and the Ollivier curvature itself carries a free idleness parameter [5, 13]; the results below are statements about *this* curvature and are not expected to transfer to the others. What motivates the geometric reading is that the Ollivier curvature of a random geometric graph converges to the Ricci curvature of the underlying manifold [14], in the density regime and for the neighbourhood radius of that theorem; the one-hop curvature used in network practice is not the object it controls, and we use it only as context. The geometric language below is therefore an interpretation of a discrete construction, not a statement about a recovered manifold. This *discrete Ricci flow* is used for community detection [15, 16], graph drawing, network comparison and the analysis of biological and infrastructural networks [17, 9, 18, 19]—the last studies a *dynamical* curvature, not the curvature-driven metric flow considered here—and sits within the broader programme of describing networks as geometric objects [20, 21].

Two of these results change the update rule or the graph. The continuous-time normalised flow based on the Ollivier–Lin–Lu–Yau curvature is well posed: solutions exist and are unique for all times [22]. Very recently Li and Münch proved that the *discrete-time* Ollivier flow $d_{n+1} = (1 - \alpha\kappa_{d_n})\, d_n$ converges to a constant-curvature metric *under the hypotheses of their theorem*—a finite, connected (irreducible) weighted graph, exact transport, and an admissible step size [23]. In the graph-embedding setting of [24] a modified flow with a contraction

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property converges; flows with *surgery*—deleting edges—converge to constant curvature on each connected component [25]; and for trees the long-time dynamics of the Lin–Lu–Yau flow have been characterised directly, including convergence of the normalised tree flow to constant curvature [26] and a caterpillar-tree characterisation of the zero-curvature limit [27]; existence, uniqueness, the directed case and sum-of-curvature identities are likewise settled [22, 28, 29].

Existing convergence results therefore concern *particular* discrete flow constructions: a specified update, an exact transport, a fixed admissible edge set, and hypotheses—connectedness above all—that a third of our corpus violates. None applies verbatim to the scheme used in the applied literature, which is multiplicative, entropy-regularised, renormalised every step, integrated for as long as the budget allows, and applied to graphs that are frequently disconnected. This paper adds no theorem. Its contribution is a measurement: a separation of four quantities that empirical work reports interchangeably (Results), and a set of controls that assign an observed residual to one of them. Only the last deserves the name "uniformisation limit", and it is not computed here:

$$V_G^* \;=\; \inf_{w\colon E\to\mathbb{R}_{>0}} \mathrm{Var}_E\big[\kappa(G,w)\big], \tag{1}$$

Nothing below is a lower bound on V_G^*. The question is not *whether* the flow uniformises—under existing hypotheses, it does—but what a finite computation can be read as saying.

What we find, in outline. The stopping rule used in empirical work is not a convergence test: 27 of the 28 runs it labelled saturated were still falling. The pooled variance on a disconnected graph is dominated by a between-component term that need not vanish, so a third of the corpus looks like a failure when it is not. Fixed-ε entropic regularisation breaks the scale invariance that would license the standard renormalisation as a gauge fixing. After all of these are controlled, connected networks from one single family still relax unusually slowly—a case study, not a demonstration that V_G^* is positive. The variance reduction we observe survives every control and is associated with the curvature–edge correspondence under the intervention we tested (8/8 families), while the structure-to-residual map does not transfer across families. Throughout, the source family, not the dataset, is the unit of independence [30, 31], and results resting on fewer than five families are labelled exploratory.

Seven controls reversed our own preliminary readings—a plateau near 0.20 that was a horizon artefact, a community effect that shrank six-fold at saturation, a spectral-gap hypothesis that failed outright, and the claim that the per-step renormalisation is a gauge fixing. We report all seven: the controls, not any single effect, are the transferable part.

Results

Analysis populations and four conflated quantities

Ten sample sizes appear below, so we fix them once (Fig. 1a): C_0, the 117 assembled datasets; C_1, the 110 that completed a first-pass run; C_2, the 93 with a complete structure/variance record—the analysis set, from 17 source families; C_3, the 80 long enough to share a common step grid; C_4, the 16 of the extended strict re-run (7 families); and C_5, the 9 re-run under exact transport (5 families).

Two distinctions organise the evidence. The first is an *evidence tier*: a result is *primary* when it rests on at least eight independent source families and *exploratory* when it rests on fewer than five, in which case its p-value is an upper bound on the evidence rather than a measurement of it. The sham control, the corpus-level component decomposition and the main variance contraction are primary; the surgery control (2 families), the transitivity manipulation (2 base graphs) and the assortativity and community sweeps (4 base graphs) are exploratory.

The second organises the paper. A number from a finite computation can be any of four

things, and empirical work reports them interchangeably:

- L1 $V(T)$: the residual at the horizon we could afford;
- L2 $V_{\mathrm{num}}(T;\theta)$: the same, under numerical choices $\theta = (\varepsilon,\ \text{solver},\ \Delta t,\ \text{normalisation})$;
- L3 the trend of $V(t)$ as the horizon T grows;
- L4 V_G^* of Eq. (1), a property of the graph alone.

Only L4 deserves the name "uniformisation limit", and $V(T) \not\Rightarrow V_G^*$: no residual reported here is a lower bound on it.

Reading a positive edge value as a length is a convention, not a measurement: in the documentation we could retrieve, none of the 28 weighted datasets is a distance—seven carry a year and one is constant. Re-reading them under three conventions moves V_0 by a median factor of 7.9, but the within-component variance still fell in all 83 completed runs (Methods, Supplementary Table S12).

The stopping criterion is not a convergence test

We first run the Ollivier–Ricci flow ($\alpha = 0.5$ idleness, Sinkhorn transport, $\dot{w} = -\kappa w$) on every network and stop when the curvature variance $\sigma^2(t)$ stops changing. The rule used in earlier studies—and in our own first pass—declares saturation when $|\sigma^2(t) - \sigma^2(t-1)|/\sigma^2(0)$ falls below a tolerance. Under it 28 runs were labelled saturated. Two statistics describe them, and they are not interchangeable: the variance fell over the final quarter in 27 of the 28 (median -24.7%, greatest -84.9%; Fig. 2d) and at the terminal step in 26 of the 28; the two that rose at the last step did so by $<0.02\%$, and one rose 1.1% over the quarter (Supplementary Table S17). Residuals reported from such runs are values along a trajectory, not asymptotes.

Suppose $V_t = V_\infty + Ae^{-\lambda t}$. Then

$$\frac{|V_t - V_{t-1}|}{V_t} = \frac{A(1-e^{-\lambda})e^{-\lambda(t-1)}}{V_\infty + Ae^{-\lambda t}}, \qquad \frac{|V_t - V_{t-1}|}{V_0} = \frac{A(1-e^{-\lambda})e^{-\lambda(t-1)}}{V_0}, \tag{2}$$

the numerators are the same and both decay, but the denominators are not. With V_0 fixed the ratio decays like $e^{-\lambda t}$ whatever V_∞ is, so the rule reports elapsed time; with V_t in the denominator it tends to $e^{\lambda} - 1$ when $V_\infty = 0$—not to zero—and to zero like $e^{-\lambda t}/V_\infty$ when $V_\infty > 0$. The direction is the one we want, but the inference is not valid in general: whenever $\lambda < \ln(1+10^{-5})$ a trajectory decaying all the way to zero also satisfies the test, and demanding 50 consecutive steps does not remove that. The corrected rule is a local diagnostic—it says the variance has stopped moving relative to itself over the observed window, nothing about the limit—and we call it an *operational stopping criterion* throughout, never a convergence criterion. It is far stricter than it looks: on the 16-network re-run of C_4 it fired once, on `kilifi` alone, the other 15 runs stopping on their step horizon or their wall-clock budget (Methods, Supplementary Table S15). `kilifi` is a disconnected network whose pooled variance is 100% between-component—one observation consistent with the argument above, not a demonstration of it. Reaching the longer horizon lowers the median residual over the same 16 networks from 0.160 to 0.109 (in 75% of them), and the residual spans five orders of magnitude, a range no single "plateau" value summarises.

The pooled estimator hides per-component uniformisation

The residual is conventionally the variance of the pooled set of edge curvatures. On a disconnected graph the flow acts independently on each component, so even if every component reaches constant curvature the pooled variance stops at the dispersion of their limiting curvatures; 34 of 93 corpus networks are disconnected, so this is not a corner case. We report two outcomes: V_{pooled} (Outcome A), retained for comparability with the literature, and V_{within} (Outcome B), the

Network	comps	steps	V_{pooled} (end)	V_{within} (end)	ratio $V_w/V_w(0)$	between share
kilifi	2	250	1.40×10^{-2}	5.83×10^{-6}	1.8×10^{-4}	100.0%
oman	5	250	1.97×10^{-2}	1.78×10^{-5}	1.4×10^{-3}	99.9%
ksa	8	183	5.80×10^{-3}	9.11×10^{-33}	—	100.0%
hw	4	65	4.29×10^{-2}	1.71×10^{-3}	1.0×10^{-1}	96.0%
g106	1	150	2.38×10^{-2}	2.38×10^{-2}	3.8×10^{-1}	0.0%
g105	1	115	1.30×10^{-2}	1.30×10^{-2}	3.1×10^{-1}	0.0%

Table 1: **Pooled versus within-component curvature variance.** For every disconnected network of C_4 the within-component variance falls by one to three orders of magnitude while the pooled variance does not. The two connected networks are shown for contrast. Horizons differ because each run had a wall-clock budget.

primary statistic for any disconnected network. Writing m_c for the number of edges in component c, V_c for the variance inside it and $\bar{\kappa}_c$ for its mean curvature,

$$V_{\text{within}} = \sum_c \frac{m_c}{m} V_c, \qquad V_{\text{between}} = \sum_c \frac{m_c}{m} (\bar{\kappa}_c - \bar{\kappa})^2, \tag{3}$$
$$V_{\text{pooled}} = V_{\text{within}} + V_{\text{between}},$$

the decomposition is exact at every weight configuration. For a connected graph $V_{\text{between}} = 0$ and the outcomes coincide, so the correction touches only the disconnected third of the corpus—but there it is decisive (Methods). Let $G = G_1 \sqcup G_2$ and suppose the flow drives each component to constant curvature, $\kappa_e = \kappa_i$ on E_i, with $\kappa_1 \neq \kappa_2$. Then $V_{\text{within}} = 0$ while

$$V_{\text{pooled}} = \frac{|E_1||E_2|}{(|E_1| + |E_2|)^2} (\kappa_1 - \kappa_2)^2, \tag{4}$$

which is positive and says nothing about whether the flow succeeded. We verified Eq. (4) numerically: for $C_5 \sqcup C_9$ at unit weights the curvatures are 0.25 and 0, the within-component variance is 0 to machine precision, and the pooled variance is 1.43495×10^{-2} against the predicted value (Supplementary Table S8). A graph can therefore be perfectly uniformised within every component and still report a nonzero pooled residual.

Across all 34 disconnected networks the within-component variance falls in 34/34, but by a median factor of only 5.7, with final values spanning 1.9×10^{-32} to 8.1×10^{-2} (Supplementary Table S3); the between-component term accounts for a median 65% of the pooled variance and more than half of it on 22 of 34. The direction is the same in all 34; the magnitude on which we earlier based the claim—three to four orders of magnitude—is the behaviour of the two most extreme networks, not the typical one, and we withdraw it.

Two of the four Pattern II networks (`kilifi`, `oman`) are disconnected, and for both the residual at the horizon is fully accounted for by this term: on those two the flow did not fail. `kilifi` is also the only run of C_4 on which the operational rule fired; `oman` stopped on its horizon (Supplementary Table S15). The two that remain—`ENZYMES_g105` and `ENZYMES_g106`—are connected, so their residual is within-component and is not an estimator artefact.

We treat these two as *case studies*: two graphs from one family out of seventeen support no prevalence claim. A third connected graph of the same family, `ENZYMES_g10`, was classified Pattern I at the 400-step horizon of the Sinkhorn re-run and is therefore not one of the two; under exact transport it also flattens, but later and from a higher residual (Section *Extended horizon*, Supplementary Table S11). We report all three on equal terms there and count all three as one family. Under every numerical control we applied—exact transport, a scale-equivariant schedule, a ten-fold range of step size, a 5,000-step horizon and two independent optimisers—they exhibited unusually slow relaxation.

Two finite-horizon relaxation patterns

Subject to that correction the remaining spread separates into two behaviours (Fig. 2e,f). Classifying each run by its decay rate over the final quarter, 11 networks show Pattern I (continued relaxation): the residual keeps falling at a median -5.5×10^{-3} per step and has reached a median of 0.024. Four show Pattern II (slow relaxation at the observed horizon): their decay rate over the final quarter falls below 10^{-4} per step and the residual sits at a median of 0.17 (0.154–0.238); one is undecided. The label describes a terminal window, not a limit.

A rate estimated in one window says nothing about the next, so Fig. 8 plots the local rate $r_t = -\log(V_t/V_{t-1})$ for all 16 runs of C_4: $r_t > 0$ is decay, $r_t \approx 0$ a plateau, $r_t < 0$ a rebound. The patterns appear as two bands; the 10^{-3} per step line between them is a descriptive threshold, not a theoretical boundary—runs cross it. What separates the patterns is the state of the weights: Pattern II networks end with a weight span of median 2.8×10^5 against 3.9×10^2 (Mann–Whitney $p = 0.039$ at network level). This is the number most exposed to clustering: the family-level bootstrap interval is $[2.64, 5.25]$ decades, but dropping any single family moves the network-level p to 0.017–0.105. We report the separation as an association that survives clustering in magnitude but not in significance, and claim no mechanism—a diverging weight span is also what makes the transport subproblem ill conditioned.

Regularisation breaks scale invariance

In Pattern II the weights span 10^5–10^{11}, so the transport cost matrices are badly scaled and the entropy-regularised Sinkhorn surrogate is at its weakest: a residual measured there could be an artefact of the regularisation strength, the solver tolerance or floating-point resolution rather than of the graph. We ran five controls; the first uncovered an error in our own framing.

Scale invariance. For the exact curvature $\kappa(G, cw) = \kappa(G, w)$, which is what would license the per-step renormalisation as removal of a null direction; for the entropy-regularised cost this fails, because with ε fixed $C \mapsto cC$ changes the kernel $\exp(-C/\varepsilon)$. Measured on nine networks and six rescaling factors (Methods), the violation reaches $\max_e |\kappa_e(cw) - \kappa_e(w)| = 1.40$ at the initial weights and 3.86 after 150 steps, while exact W_1 and a scale-*equivariant* schedule $\varepsilon_t \propto s_t$ are invariant to machine precision. The claim in our previous version—that the renormalisation removes a null direction—is false as stated and we withdraw it; $\sum_e \log w_e = 0$ below is a normalisation of the search space, not a gauge fixing.

Two things bound the damage: s_t drifts by only 0.98–1.29× over 200 steps, so the violation does not compound; and repairing it barely changes the numbers. Under the scale-equivariant schedule at a matched horizon of 200 steps the pooled residual moves by a median −4.6% (at most −14%) with no change of ordering, and replacing regularised transport by exact transport *raises* the residual on the two connected networks (+5% on `g105`, +193% on `g106`)—slower relaxation, not a lower floor.

Exact transport at a long horizon. Running population C_5 with exact W_1 transport for 2,000 steps separates the networks cleanly (Fig. 3c; Methods gives the values): five reach a within-component variance below 10^{-11}—two connected and three disconnected, the latter with a pooled variance of 0.033–0.112 that is entirely between-component—and four do not, among them the three connected ENZYMES graphs. Both halves of the story survive exact transport.

Extended horizon. Extending the three connected ENZYMES runs from 2,000 to 5,000 steps under exact transport changes their within-component residual by +0.009% (`g105`), +0.0002% (`g106`) and +0.17% (`g10`); in absolute terms 1.1×10^{-6}, 6.8×10^{-8} and 8.7×10^{-5}, while the weight span grows to 10^{65}–10^{72} (Supplementary Table S11). All three satisfy the descriptive criterion used elsewhere in the paper—a relative per-step move of the variance below 10^{-5} sustained for at least 50 consecutive steps—for 3,920–4,241 consecutive steps, and on 79–85% of all 5,000 steps (Supplementary Table S16); they are the only runs here on which it is met. What follows is a statement about the 2,000–5,000-step window we observed and nothing about

$t \to \infty$. We report the three together and count them as one family, not as three independent observations.

They are not interchangeable, which is why two of them carry the case study: `g10` is still relaxing at step 1,000 (its residual falls 14.3% between steps 1,000 and 1,500), so its plateau begins later and at a value (5.2×10^{-2}) not comparable with a 2,000-step reading of the other two, and its terminal drift ($r_t = -1.4 \times 10^{-7}$ per step, against -1.3×10^{-10} and -2.4×10^{-13}) is larger but still three orders of magnitude below the 10^{-3} line. In all three the sign of r_t at the end of the run is negative: the variance is rising, not falling, and the residual at the horizon lies above the minimum reached earlier in the same run by 0.5% (`g10`), 2.8% (`g105`) and 11% (`g106`). A finite-horizon residual is a value on a non-monotone trajectory, not a statement about the infimum.

Step size at fixed physical time. Holding physical time fixed at $T = 20$ and varying Δt ten-fold, from 0.01 (2,000 steps) to 0.1 (200 steps) under exact transport on C_5, the pooled residual at T changes by at most 0.033% on the seven networks that have not reached the operational uniformisation threshold by then, and by up to 4.9% on the two that have (`aves-weaver-social-14`, $8.0 \times 10^{-6} \to 7.8 \times 10^{-6}$, and `scc_rt_http`, $1.06 \times 10^{-5} \to 1.01 \times 10^{-5}$; Supplementary Table S7). Both of the larger changes are on residuals already at the uniformisation threshold, so 4.9% of 10^{-5} is 5×10^{-7} in absolute units. The same caveat applies to `scc_rt_oman` ($7.1 \times 10^{-8} \to 5.8 \times 10^{-8}$: 18.6% in relative terms, 1.3×10^{-8} in absolute terms). What we can claim is therefore that the residual is *largely stable to step-size variation at matched physical time for the tested non-converged trajectories*. That is weaker than step-size convergence—one physical time, one idleness, one initial weighting and nine networks were tested—and we do not claim the latter.

Regularisation strength. Entropic smoothing shrinks curvature magnitudes, and could hide or invent a spread. Re-run at a *matched* horizon over a 20-fold reduction of ε with periodic exact-transport checkpoints, the exact-transport residual spans 5.1% on `g105` and 9.7% on `g106`; where the estimators disagree, Sinkhorn is *below* exact (3.6× on `g106`), so regularisation hides rather than creates the residual. At $\varepsilon = 0.002$ the solve fails and those runs are excluded (Supplementary Table S9).

No observed residual is claimed to bound V_G^*.

The plateau tracks the curvature–edge map

A plateau could, in principle, be a mechanical artefact of the update rule $w \leftarrow w\, e^{-\Delta t\, \kappa}$ with renormalisation, rather than a property of the geometry. We therefore ran a *sham flow*: identical numerics, but at every step the curvature values are randomly permuted across edges before the update (Methods). If homogenisation were mechanical, the sham arms would homogenise too. They do not. These 25 pairs are not 25 independent experiments: they come from 8 source families. At the network level the split is 24/25 (Methods); aggregating within family and testing across families, all 8/8 families show the sham arm higher (sign test $p = 3.9 \times 10^{-3}$; family-level sign-flip permutation $p = 3.9 \times 10^{-3}$; cluster-bootstrap 95% interval for the median log-ratio gap $[-2.60, -0.69]$, excluding zero), and the worst leave-one-family-out p is 9.2×10^{-4} (Fig. 2c, Methods). The effect size is unchanged (a median factor of 2.6), but the evidence is that of 8 clusters, not 25 networks. Destroying the curvature–edge mapping does not merely slow the flow; it suppresses homogenisation over the tested horizon. The homogenisation we observe is therefore *associated with* the geometric correspondence between edge curvature and edge update, and is not reproduced by the update rule alone when that correspondence is randomly destroyed. The control randomises the pairing and the spatial arrangement together, so it shows the correspondence matters under this intervention, not that no other mechanism could.

Relaxing the edge-set constraint

Surgery-based discrete Ricci flows converge to constant curvature by deleting edges [25]. Our counterpart control asks the converse question: how much of the residual is due to edges the flow is not allowed to remove? In paired runs with identical numerics, removing only the edges whose curvature stays above a fixed tolerance (4–14 edges of 41–84, with bridge edges protected) and re-running the flow collapses the residual in 5/5 defensible pairs (one-sided sign test $p = 0.031$; Fig. 4), by a median factor of 6.6 and up to 132×. The five pairs come from only two families (four ENZYMES, one `scc_rt`), so the family-level test has essentially no power: both families show the relaxed arm lower, but with two clusters a sign test cannot go below $p = 0.25$. The direction is consistent in every family (median drop 25× within ENZYMES, 1.05× for the single `scc_rt` pair); the magnitude is not established at family level, and the network-level $p = 0.031$ should be read as an upper bound on the evidence.

This is a *constraint-relaxation* control, not a causal demonstration: the intervention is outcome-informed and bridges are protected by fiat. The defensible claim is that the residual measures how far a graph is from constant curvature *if its edge set may not change*—which makes our result complementary to, rather than in conflict with, the surgery literature, whose flows destroy exactly this quantity.

Transitivity changes the relaxation time

Why do some networks homogenise almost completely while others retain a large residual at the same horizon? We manipulated single structural quantities on synthetic graphs while holding the degree sequence exactly fixed, using degree-preserving double-edge swaps [32] (Methods). Raising transitivity from 0.12 to 0.43–0.46—the transitivity of the median real network—raises the residual at a matched horizon of 250 steps by three orders of magnitude on both base graphs. Followed to terminal states, both arms of the first graph reach the operational uniformisation threshold, at step 381 (low transitivity) and 884 (high), and the fold between them collapses from 31× at 125 steps to 1.02 at 900 (Fig. 5b,c). The effect is therefore entirely a difference in time scale: the arms end in the same state and differ only in arrival time (2.3× in steps), because the high-transitivity arm decays slowly then accelerates 11-fold. Nothing measured in one window predicts the next. Within the tested design, transitivity was the structural variable most clearly associated with the observed relaxation-time difference, while the assortativity change carried along with it accounts for only 2.5–3.4% of it. Two earlier readings are withdrawn: that transitivity raises the *plateau* by 25–37×, and that it selects a relaxation pattern. Both were horizon artefacts. The manipulation rests on two base graphs and is *exploratory*. The assortativity and community sweeps (Methods) are secondary: both ran under the unsafe stopping rule, and at saturation the community effect shrinks from 6.1× to 1.35×.

The structure-to-residual map does not transfer

Transitivity and assortativity correlate with the residual within bases and within the largest family in our corpus (ENZYMES, $n = 15$ at the operational threshold: $\rho = +0.73$ for transitivity, $p = 0.002$). But the correlation is *not* a predictor. In leave-one-family-out cross-validation across 25 networks, only assortativity rises even marginally above the intercept baseline, and it still explains essentially none of the variance (Fig. 6a): assortativity reaches out-of-sample $R^2 = +0.04$ against a baseline of -0.17, modularity reaches -0.11, and the four-variable model -0.79. Within ENZYMES the transitivity correlation collapses once size is controlled (partial $\rho = +0.25$, $p = 0.37$, with size itself at $\rho = -0.76$), and at the family level the trend is positive but not significant (5 families, $\rho = +0.50$, $p = 0.39$; Fig. 6b). The relation between global structure and the residual is *ecological*: it holds between families rather than within them, and it does not transfer. This is a null result and we regard it as one of the main findings. An out-of-sample R^2 below the intercept baseline is evidence that *these models do not transfer*, not

that the residual is intrinsically family-specific; the latter would require showing that no statistic predicts it. Of the descriptive correlations reported alongside the test, assortativity ($\rho = +0.41$) survives family-level resampling (95% interval $[+0.18, +0.72]$) and transitivity ($\rho = +0.22$) does not ($[-0.30, +0.55]$; Fig. 6c): in-sample associations should be read with the family, not the network, as the unit.

No evidence that the spectral gap predicts decay

Finally, the dynamics. Curvature controls eigenvalue estimates [6], so the spectral gap is a natural candidate predictor. The early decay rate of the curvature variance is uncorrelated with the spectral gap of the initial Laplacian across 41 usable units spanning three orders of magnitude in λ_1 ($\rho = -0.03$; $|\rho| \geq 0.31$ detectable at $\alpha = 0.05$), and in a joint model the gap carries no information beyond size and mean degree ($\beta = -0.10$, $p = 0.18$), while mean degree dominates ($\beta = +0.70$, $p = 3 \times 10^{-4}$; Fig. 7a). This null is robust to clustering: over the 11 families the cluster-bootstrap interval for ρ is $[-0.43, +0.39]$, and the sign is stable in only 58% of replicates. We report this as the negative result it is: in this sample we find no evidence that the initial spectral gap predicts the observed early decay rate. It does not follow that mean degree is the causal driver—a regression coefficient on observational network data is not an intervention. What the rate tracks is where the flow is at the end of the run ($\rho = -0.43$; partial $\rho = -0.63$ controlling for size and mean degree; Fig. 7b). At the network level this is significant ($p = 1 \times 10^{-5}$), but the family-level interval is $[-0.68, +0.16]$ and includes zero, so we withdraw our earlier formulation of it as a result. Part of the association is definitional: the fitted rate and the final variance summarise the same trajectory.

Discussion

Four statements survive the controls above, in order of evidential strength, each at the level its evidence supports.

1. A finite-time residual is not a convergence certificate. The stopping rule used in empirical work divides the per-step change of the variance by its initial value; Eq. (2) shows that this ratio decays like $e^{-\lambda t}$ whatever the limit is, so the rule reports elapsed time—27 of the 28 runs it labelled saturated were still falling. The *operational stopping criterion* used instead says only that the variance has stopped moving relative to itself over the observed window.

2. On a disconnected graph the pooled curvature variance can substantially overstate within-component non-uniformity. Eq. (4) exhibits the case $V_{\text{within}} = 0$ with $V_{\text{pooled}} > 0$: a graph can be perfectly uniformised within every component and still report a nonzero residual. The between-component term is a function of the component mean curvatures, which themselves evolve, so we do not claim that it cannot decay—only that it need not, and that on this corpus it does not. Across all 34 disconnected networks of the analysis set it falls by a median factor of 5.7; our earlier claim of three to four orders of magnitude is withdrawn. A third of the corpus is disconnected and the pooled variance is the standard reporting choice, so the correction applies to previous work too.

3. Entropic regularisation changes the numerical curvature dynamics unless its scale is controlled. With ε held fixed the regularised curvature is not invariant under $w \mapsto cw$, so the reading of the per-step renormalisation as a gauge fixing is unavailable and we withdraw it. Exact transport and a scale-equivariant schedule restore invariance, and the residual survives both: at a matched horizon it moves by $\leq 9.7\%$ over a 20-fold ε reduction and by $\leq 4.9\%$ over a ten-fold Δt range at $T = 20$. We claim that the residual is not explained by the tested controls; we do not claim immunity to every regularisation, conditioning or precision effect.

4. After these controls some connected networks still relax very slowly, but the evidence does not establish a positive geometric infimum. Three connected ENZYMES networks—three graphs

Hypothesis / ingredient	Li & Münch [23]	Scheme used here
Graph	finite, weighted, connected (irreducible)	finite, weighted; 34 of 93 disconnected
Curvature	exact Ollivier W_1	entropy-regularised Sinkhorn, $\varepsilon = 0.05$
Update	$d_{n+1} = (1-\alpha\kappa)d_n$	$w_e \leftarrow w_e e^{-\Delta t\,\kappa_e}$, $\Delta t = 0.05$
Renormalisation	not part of the statement	mean weight renormalised every step
Step size	admissible range	fixed; $\alpha = 0.5$ and 0.05 both tested
Edge set	fixed, no surgery	fixed; surgery arm adds a control only

Table 2: **Why the convergence theorem does not transfer.** Each row is an ingredient on which the scheme studied differs from the one proved to converge.

from one family of seventeen—changed their within-component residual by +0.009%, +0.0002% and +0.17% over 3,000 further steps of exact transport, to a 5,000-step horizon, while their weight span grew to 10^{65}–10^{72}; at those weights the residual survives every numerical diagnostic we applied (Methods). Two of the three carry the full control battery; the third flattened later and from a higher residual. A metastable transient longer than any horizon we could afford fits the data equally well.

The variance reduction we observe is associated with the curvature–edge correspondence under the intervention we tested: destroying it in the sham control is associated with suppressed homogenisation in 8/8 families. One intervention in one finite window is evidence of association, not a proof that the correspondence is necessary, nor that every network or flow scheme must homogenise. What remains unpredicted is which real networks relax quickly: the structure-to-residual map does not transfer across families, transitivity moves the relaxation time and not the endpoint *in the synthetic design we tested*, and a rate measured in one window carries no information about the next, which is why Fig. 8 reports the rate trajectory rather than a classification.

Table 2 sets out, hypothesis by hypothesis, why the convergence theorem most relevant to us does not transfer to the scheme studied here—a statement about the hypotheses, not a criticism of the theorem. Where the hypotheses hold we observe what the theorem predicts; the remaining differences—exponential versus Euler update, regularised versus exact transport—we tested directly (Methods), and neither produces the residual.

For practitioners: stop when the variance stops moving relative to *itself*; report the within-component variance whenever the graph is disconnected; with a fixed-ε solver report ε/s_t or use a scale-equivariant schedule; treat neither the residual as predictable across families nor same-source networks as independent replicates.

Methods

Corpus. We assembled 117 weighted-network datasets from a public network repository, covering 12 domains (cheminformatics, animal social networks, dynamic contact networks, temporal reachability, biological, brain, power, interaction, web, ecology, infrastructure and economic networks). 110 runs completed. The analysis set comprises the 93 networks with a complete structure/variance record; 17 runs were excluded (degenerate initial curvature variance at float-noise level, or runs that could not start/complete for size or load reasons). Within it, 41 runs terminate by the operational stopping criterion defined below: 34 reached the current-variance threshold ("saturated") and 7 reached their step horizon, so these 41 form the converged subset; a stricter tail-drift test leaves 31. These counts describe how each run *ended*, not convergence: the earlier first-pass rule (denominator $\sigma^2(0)$) had labelled 28 runs saturated even though 27 of them were still falling (Results; Supplementary Table S17). The analysis set spans $n = 4$ to 6,809 nodes (median 52) and 17 source families (11 with ≥ 3 networks); 34 of 93 networks are disconnected. Supplementary Table S1 lists every dataset with its exclusion status.

Flow. Ollivier–Ricci curvature with idleness $\alpha = 0.5$, optimal transport computed by entropic-regularised Sinkhorn iterations (regularisation $\varepsilon = 0.05$) [33, 34, 35, 36]. Each step applies $w_e \leftarrow w_e \exp(-\Delta t\, \kappa_e)$ with $\Delta t = 0.05$, followed by renormalisation of the mean weight to 1. Edge weights act as edge lengths. *This is a normalised, entropy-regularised discrete Ricci flow, not an exactly scale-invariant Ollivier–Ricci flow*: with ε held fixed the curvature is not invariant under $w \mapsto cw$ (Results), so the per-step renormalisation is a constraint on the state and not the removal of a null direction, and no result here should be read as a statement about the exact flow unless it is explicitly derived from the exact-transport re-runs described below. Three variants of the same scheme are run as controls: *exact transport*, in which every transport problem is solved by network simplex (POT `ot.emd`, with a HiGHS linear-programming fallback) so that $\kappa_e = 1 - W_{1,e}/d_e$ is the exact Ollivier curvature; *scale-equivariant regularisation*, in which $\varepsilon_t = \varepsilon_0\, s_t/s_0$ with s_t the mean entry of the transport cost matrices at step t, which restores exact equivariance; and the *first-order Euler update* $d_e \leftarrow (1 - \alpha\kappa_e)\, d_e$ studied by [23], run with and without the mean-weight renormalisation and with $\alpha \in \{0.05, 0.5\}$, so that the only difference from the production scheme is the update rule. *Initial weights.* Where a dataset supplies finite, strictly positive edge values those values are used as the initial edge lengths (`weight_mode = auto` with the `distance` convention); where a dataset is unweighted or carries non-positive values—the majority of the corpus—every edge is initialised to unit length. "Weighted network" in this paper therefore refers to the *evolving* weight function, not to an assumption that the source data carry physical weights. Reading a positive third column as a length is a *modelling choice*: in public repositories such a value usually encodes an affinity, a count, a flow or a timestamp, and we hold no verified data dictionary for any of the 28 weighted datasets. We therefore do not assert a physical meaning for any of them; Supplementary Table S12 lists every weighted dataset with the observed range and type of its third column, and the control below re-runs them under two alternative readings. Runs use a step horizon and a wall-clock budget, whichever is reached first; per-step trajectories are stored.

Two experimental stages. The corpus screening (110 datasets, 10 steps, $t = 0.5$) and the long-run study (16 networks re-run to 400–912 steps) are separate experiments and are reported as such. The 10-step stage establishes only that curvature heterogeneity falls in the transient; every statement about long-time behaviour comes from the second stage. In particular, the first stage's 10-step horizon sits at $V = 0.781$ of the initial variance on the median trajectory, which is a transient, not a result.

Convergence. Two stopping rules are used, and the difference between them is one of the results. The *initial-variance* rule, used for the first corpus pass and in earlier studies, stops when $|\sigma^2(t) - \sigma^2(t-1)|/\sigma^2(0) < 10^{-4}$ for a fixed window; its denominator is constant while the per-step change decays, so any run whose increment tends to zero may eventually satisfy it—exponential decay is the textbook case. Of the 28 runs it labelled saturated, 27 were still falling over the final quarter and 26 at the terminal step (median quarter change −24.7%; Supplementary Table S17). The *current-variance* rule configured for the strict re-run stops when $|\sigma^2(t) - \sigma^2(t-1)|/\sigma^2(t) < 10^{-5}$ for 50 consecutive steps, evaluated on the pooled variance, which is the quantity the integration loop monitors. We recomputed the test step by step from the recorded trajectories (Supplementary Table S15). It fired on 1 of the 16 runs—`kilifi`, whose final 50 steps are the only 50 steps of its trajectory below the tolerance; the run ended there, at step 345. The other 15 terminated on their step horizon (245–400 steps) or on the 700 s wall-clock budget, and on 11 of the 15 no step of the trajectory came within the tolerance at all; the remaining four satisfy it intermittently but never for 50 consecutive steps (longest streak 1–33). For those 15 the smallest tolerance at which the rule would have fired somewhere in the final fifty steps has median 5.1×10^{-3} per step (range 4.2×10^{-5}–1.7×10^{-2}), about 500× the configured tolerance on the median run; the three 5,000-step exact-transport runs satisfy the same test with maximal per-step relative moves of 1.3×10^{-7}, 8.9×10^{-11} and 1.1×10^{-13} (Supplementary Table S16). The terminal-window statistics below are therefore not in conflict with the rule; they are the record of runs on which it did not fire. Over the final 50 steps of the 16 strict runs, the per-run median of r_t has median 4.7×10^{-3} per step across runs and ranges from -3.3×10^{-5} to 1.7×10^{-2}; 14 of 16 runs have a terminal median rate above 10^{-4} per step, and—the observation that most directly falsifies any forward extrapolation—two runs have a *negative* terminal median rate, i.e. their variance was rising at the end of the run. The realised relative change of V over the same final window is 21% in magnitude on the median run and up to 58%. A small change over fifty steps is therefore a description of those fifty steps, not evidence about the next fifty. No run of C_4 is described as converged, and we use only three termination labels: *rule-terminated* (the operational rule fired; 1 of 16), *horizon-terminated* or *budget-terminated* (the run stopped on its step horizon or its wall-clock budget; the remaining 15), and *still falling at the horizon* (terminal median $r_t > 10^{-4}$ per step; 14 of 16). The string *near-plateau* is used only for the three 5,000-step exact-transport runs of Supplementary Table S16, where the criterion is verified to hold; it is never inferred from a run that ended on its horizon.

Two qualifications accompany the rule. First, monotonicity of the variance is *assumed*, not proven: we verified empirically that 10 of the 16 strict trajectories never increase; that the remaining six do so on at most 16.3% of steps (234 of 6,052 increments) with a maximum single increase of 0.39% of the current value; and that the cumulative upward movement is at most 3.7% of the total decrease achieved. Because the criterion fires on $|\Delta V|$ rather than on its sign, these violations do not affect any regime assignment, but the last observed value of a truncated run is treated as a finite-horizon residual, not as a rigorous upper bound on the asymptotic value. Second, as $V \to 0$ the relative criterion's denominator also tends to zero, so we additionally monitor the absolute increment $|\sigma^2(t) - \sigma^2(t-1)|$; every run reported as having reached the operational uniformisation threshold ($V \leq 10^{-6}$ of the initial variance) also has an absolute increment below 10^{-9} in variance units, i.e. it is far from the regime where the relative test is numerically vacuous. Sixteen networks were re-run under the strict rule; per-network step counts, residuals, decay rates and weight spreads are given in Supplementary Table S2.

Variance decomposition. For every run we record the component label of each edge and decompose the pooled curvature variance as in Eq. (3), where c indexes connected components, m_c is the number of edges in component c and $\bar{\kappa}_c$ its mean curvature. Both terms are evaluated at the same weight configuration as the pooled value, so the decomposition is exact. For a connected graph the decomposition is trivial ($V_{\text{within}} = V_{\text{pooled}}$); we therefore re-ran the disconnected

networks of the analysis set to the horizon each reached in the first pass (capped at 200 steps and 900 s of wall clock). All 34 completed, and both outcomes are reported for every one of them (Supplementary Table S3). The corpus-level statistics quoted in this paper use V_{pooled} for connected networks and V_{within} for disconnected ones.

Scale-invariance probe. On nine networks (the four Pattern II runs and five controls) we evaluate the curvature at the current weights, then multiply every edge weight by $c \in \{0.25, 0.5, 2, 4, 10, 100\}$ and re-evaluate, reporting $\max_e |\kappa_e(cw) - \kappa_e(w)|$ and the change in $\mathrm{Var}[\kappa]$. The probe is run at the initial weights and after 150 flow steps, and separately for the three transport backends (fixed-ε Sinkhorn, scale-equivariant Sinkhorn, exact W_1). The cost scale s_t is recorded at every step of every run, which is what makes the drift of the effective ratio ε/s_t measurable (Supplementary Table S6).

Numerical controls. *Further numerical controls.* Re-running the flow with $\varepsilon \in \{0.1, 0.05, 0.02, 0.01\}$ leaves the residual on `kilifi` invariant to 0.9% (0.1771–0.1788 at 250 steps) and *raises* it on the other three (`oman` 0.170 → 0.235, `g106` 0.175 → 0.529, `ksa` 0.146 → 0.964), in each case with a smaller weight span at the same horizon: slower relaxation, not a higher plateau. The residual does not vanish as the regularisation is removed. This is not simply the finite-ε bias of the transport cost relative to exact W_1, which is a separate and well-studied quantity [37, 38, 39, 40]: debiased formulations, stabilised ε-scaling and sparse exact schemes address different aspects of that bias, and we do not claim to have removed it. What we observe is a property of the trajectory—the horizon at which the residual is read moves with ε. Tightening the solver from $\mathrm{tol} = 10^{-9}$ to 10^{-13} and from 5×10^3 to 5×10^4 iterations moves the variance by 3×10^{-5} to 2×10^{-2}. Two searches of a different algorithmic class from the flow (a trust-region solver for the uniformisation equations $\kappa_e(w) - \bar{\kappa}(w) = 0$, and multi-start quasi-Newton minimisation of $\mathrm{Var}[\kappa]$ over $\log w$) found, under a matched wall-clock budget, no weighting of lower variance than the flow on five of six networks; on `kilifi` they agreed with it to 2% (0.177, 0.180) and neither solved the equations ($\max_e |\kappa_e - \bar{\kappa}| = 1.3 \times 10^{-2}$). No weighting with substantially lower variance was found; we make no claim that any observed residual bounds V_G^*. (Full descriptions are in Methods.)

Numerical controls. *Regularisation sweep:* the flow is re-run on both Pattern II networks (`g105`, `g106`) with $\varepsilon \in \{0.1, 0.05, 0.02, 0.01, 0.005, 0.002\}$, everything else held fixed, with the curvature re-scored by exact transport at periodic checkpoints so that the Sinkhorn-to-exact discrepancy $\max_e |\kappa_e^{\text{sinkhorn}} - \kappa_e^{\text{exact}}|$ is recorded along the trajectory. Crucially the runs are compared at a *matched* step index, not at their own stopping step, because the stopping step depends on ε: 239 steps on `g105` and 400 on `g106`, the largest horizon common to all six values. On `g105` the residual at $t = 11.95$ is 9.8×10^{-3} to 1.0×10^{-2} for ε from 0.1 down to 0.005 (a spread of 5.5%), and the exact-transport value on the same weights is 1.14×10^{-2} to 1.20×10^{-2} (5.1%). On `g106` at $t = 20$ the exact-transport value is 3.44×10^{-2} to 3.78×10^{-2} (9.7%), while the Sinkhorn value at $\varepsilon \geq 0.02$ is 1.03×10^{-2}, i.e. 3.6× *lower*: entropic smoothing shrinks curvature magnitudes, so the regularisation understates the residual and cannot be its source. As ε falls the Sinkhorn value rises toward the exact one ($\max_e |\Delta\kappa_e|$ falls from 0.69 to 0.095 between $\varepsilon = 0.02$ and 0.01). At $\varepsilon = 0.002$ the solve itself degrades: $\max_e |\Delta\kappa_e|$ reaches 29 on `g105` and 4.44 on `g106` at the matched checkpoint, and on `g106` even the initial variance is wrong by a factor of 2.5 (0.025 against 0.063 at every other value of ε), so both runs are excluded rather than interpreted. *Precision probe:* the final weights of each sweep are re-scored with the transport solver tightened from $\mathrm{tol} = 10^{-9}$ to 10^{-13} and from 5×10^3 to 5×10^4 iterations. *Independent optimisation:* two searches of a different algorithmic class from the flow are run over $\theta = \log w \in \mathbb{R}^{|E|}$. Under the exact curvature the objective would be invariant under adding a constant to θ, and $\sum_e \theta_e = 0$ would be a gauge fixing; under the fixed-ε regularised curvature

used here it is not, so $\sum_e \theta_e = 0$ is imposed as a *normalisation of the search space* and the search is re-scored with the production solver throughout. The first solves the uniformisation equations $F_e(\theta) = \kappa_e(e^\theta) - \overline{\kappa}(e^\theta) = 0$ with a bounded trust-region least-squares solver; the second minimises $\mathrm{Var}[\kappa]$ by multi-start quasi-Newton with finite-difference gradients and a derivative-free polish. Both use a relaxed solver during the search and re-score every candidate optimum with the production solver settings, so the comparison with the flow is exact. Search dimension ranges from 13 to 69 and each network is given a fixed wall-clock budget. Under that budget neither search found a lower-variance solution than the flow on five of the six networks. This compares algorithmic efficiency at matched cost, not global optimisation ability, and we report it as a property of the search as well as of the problem.

Weight-semantics sensitivity. Twenty-eight of the 117 assembled datasets carry a finite, strictly positive third column and were therefore initialised with it as the edge length. To test how much of the result depends on that reading, each of them is re-run under three initial conventions with identical numerics and a matched step horizon: *as-is* ($l_e = w_e$, the production choice), *reciprocal* ($l_e = 1/w_e$, the reading appropriate if the value is an affinity, a count or a flow) and *unit* ($l_e = 1$, topology only). The three conventions give the same topology and the same flow, and differ only in the initial metric. For each network we compare the initial curvature variance V_0, the residual at the matched horizon and the late-window decay rate. The outcome is summarised in Results and given per dataset in Supplementary Table S12. No convention is designated correct: the control bounds how much of the corpus result is a property of the data rather than of the reading we gave them. All twenty-eight datasets were run under all three conventions, giving 84 convention×dataset runs of which 83 completed (one aborted on a numerical overflow, `eco-mangwet` under the reciprocal convention; Supplementary Table S12). Eight of the twenty-eight have a raw third column with a coefficient of variation below 2×10^{-3}—the seven `reptilia-tortoise-network-*` graphs, whose entries run from 1996 to 2015 with two to ten distinct values, and `bio-DM-LC`, whose column is constant—so for them the three conventions differ in the input metric by less than one part in five hundred and the control is close to vacuous: for these eight the measured V_0 agrees across conventions to within 0.05%. On the 20 datasets with a genuinely varying column the conventions are different inputs and V_0 differs across them by a median factor of 7.9 (range 1.0–1.2×10^{13}, the largest on the heaviest-tailed column, `eco-mangwet`). The qualitative outcome does not move with the reading: the within-component variance at the horizon lies below its initial value in all 83 completed convention×dataset runs (the one aborted run excluded), with medians 0.63 (as-is), 0.52 (reciprocal) and 0.83 (unit), and a largest ratio of 0.99. Many of these runs are short, so the comparison was repeated on the 30 runs that reached at least 20 steps: the variance again lies below its initial value in 30 of 30, with medians 0.10, 0.13 and 0.49 and a largest ratio of 0.72. Horizons are not equal across datasets—each run had a fixed wall-clock budget—so these ratios compare conventions within a dataset, not datasets with each other. Only 3 datasets reached the 250-step target under all three conventions, and they agree to three significant figures (`hw` 0.1751, 0.1745, 0.1747; `mc` 0.3897, 0.3898, 0.3897; `sg` 0.007589, 0.007595, 0.007591); on every other dataset the budget cut at least one convention short, so those comparisons are not step-matched. One run, `eco-mangwet` (recip), aborted on the overflow guard because the reciprocal of a column spanning nine decades produces lengths whose curvatures are large enough to overflow the update; it is excluded rather than repaired. All 28 of the 28 datasets therefore contribute all three convention runs except this single aborted one.

Sham flow. Identical numerics, but at each step the vector of curvature values is randomly permuted across edges before the weight update. Both arms are step-matched (identical step counts). The exclusion used in our first analysis—real arm inert (ratio > 0.8) or sham arm divergent (ratio > 5)—is an exclusion on the *outcome*, so we report both analyses. Over

all 25 pairs the real arm is lower in 24/25 (one-sided sign test $p = 7.7 \times 10^{-7}$; Wilcoxon $p = 8.9 \times 10^{-8}$; median real residual 0.127 against sham 1.147); over the 21 pairs surviving the pre-specified exclusion, 21/21 ($p = 4.8 \times 10^{-7}$; medians 0.079 and 1.172). One network reverses (`bn-macaque-rhesus_interareal`, real 1.0001 against sham 0.9919); it is one of the two networks whose real arm is inert, and it is the single exception in either analysis. The conclusion does not depend on the exclusion, and we report the all-pairs result as the primary one. Each pair uses one permutation realisation; inference is by paired sign test across independent networks rather than by replicates within a network, so we characterise the direction and effect size of the sham–real difference but not a per-network sham distribution.

Constraint-relaxation control. Edges whose curvature remains above a tolerance are removed (bridge edges protected to preserve connectivity), and the flow is re-run from the rewired state with identical numerics. A pair is *defensible* when at least 3 edges are removed and the surgery arm completes; smaller removals are perturbations, not surgery, and are excluded. Five pairs qualify. The intervention is outcome-informed—edges are selected using the curvature the flow produced—so we treat it as a constraint-relaxation control and not as a causal experiment.

Synthetic manipulations. Structural quantities follow their standard definitions: transitivity (clustering coefficient) [41], degree assortativity [42] and modularity [43]. Starting from base graphs, degree- preserving double-edge swaps are accepted/rejected to drive a single target statistic (assortativity, transitivity, or modularity) to a requested value, holding the degree sequence exactly fixed (verified by hashing). Each cell is run to saturation; saturation status is recorded per cell. The assortativity–transitivity–modularity triangle is collinear in this design, which is why each quantity is manipulated separately and effects are attributed through the calibrated fixed-effects model.

Long runs. Extending a run beyond the point at which a single process can be relied upon is done with a checkpointed integrator that saves the edge weights, the step counter and the stopping-rule state after every chunk and resumes from them. The continuation is not an approximation: it uses the same update, the same configuration and the same seeded graph, and we verified that it reproduces the original trajectory to the last digit ($V = 0.047023126080126196$ at step 249 in both). Each continuation also writes the per-step variance, which is what makes the window-by-window decay rates in Fig. 5 computable.

Statistics. We use an explicit four-level independence hierarchy:

$$\text{family (source)} \;\rightarrow\; \text{dataset} \;\rightarrow\; \text{trajectory} \;\rightarrow\; \text{time step}$$

in which level 1 is the unit of independence, level 2 the experimental unit, level 3 a repeated experiment and level 4 a repeated measure. Trajectory steps are never treated as independent observations, and no test in this paper uses steps as replicates. Correlations are Spearman; group comparisons use one-sided exact sign tests with Wilcoxon confirmation.

Declaring level 1 to be the unit of independence and then testing at level 2 would be anti-conservative, since datasets that share a source share a generator, a file format and a sampling bias [30, 31]. Every headline p-value is therefore reported twice: once at the network level (the conventional, anti-conservative reading) and once under family clustering, using three procedures that agree wherever all three are computable—a *family-level sign test*, in which the per-family median of the paired difference is the observation and the sign is tested across families; a *family permutation test*, in which whole families are sign-flipped together (computed exactly when the number of families is ≤ 20); and a *cluster bootstrap*, in which families are resampled with replacement (4,000 replicates, percentile intervals). Of the three, the exact family-level permutation and the sign test carry the inferential weight: with 8–17 clusters the percentile

bootstrap is reported as a sensitivity analysis and is never the primary test [30, 31]. We also report the leave-one-family-out jackknife, i.e. the worst p-value obtained after dropping any single family, which is the most direct answer to "does one family drive this?". Effect sizes are unchanged by all of this; only the error bars are. Where the number of clusters is too small for a test to have power (the constraint-relaxation control has two families, the synthetic sweeps four base graphs), we say so and report the direction and the network-level p as an upper bound on the evidence rather than pretending the clustered test is informative.

Out-of-sample prediction uses leave-one-family-out cross-validation with the intercept-only model as baseline, so a network is always predicted from other families. For the synthetic sweeps the corresponding hierarchy is base graph → rewiring realisation → target value: attribution is made with base-graph fixed effects, and the clusters for the bootstrap are the four base graphs. Sign flips between the full sample and the subset that reached the operational threshold are reported rather than resolved by re-thresholding. Where an analysis depends on an exclusion that is itself an outcome, we report the analysis both with and without it.

Detailed numerical results for the controls reported in Results. In the strict sample the within-component variance falls from 3.15×10^{-2} to 5.83×10^{-6} on `kilifi` (2 components; 99.96% of the pooled residual between-component), by a factor of 722 to 1.78×10^{-5} on `oman` (5 components, 99.9%), and to 9×10^{-33} on `ksa` (8 components, 100%): each component reaches constant curvature to machine precision and the pooled value is entirely their disagreement. The same holds, to a lesser degree, for `reptilia-tortoise-network-hw` (96.0%). Scale-invariance probe (detail): nine networks of population C_5, rescaling factors $c \in \{0.25, 0.5, 2, 4, 10, 100\}$, evaluated at the initial weights and after 150 flow steps. Maximum curvature change under exact W_1: $\leq 4 \times 10^{-16}$; under the scale-equivariant schedule: $\leq 1 \times 10^{-15}$ (a theorem, since $\exp(-cC/c\varepsilon) = \exp(-C/\varepsilon)$); under fixed $\varepsilon = 0.05$: 1.40 (initial) and 3.86 (after 150 steps), with the curvature variance moving by up to 0.59. The mean entry s_t of the cost matrices drifts by only 0.98–1.29× over 200 steps, so the violation does not compound. Exact transport to 2,000 steps (detail, population C_5): within-component variance at the horizon—`aves-weaver-social-14` 9.2×10^{-12} and `scc_rt_http` 3.2×10^{-18} (connected); `kilifi` 4.2×10^{-13}, `oman` 3.2×10^{-31} and `ksa` 8.0×10^{-33} (disconnected, with pooled variances 0.054, 0.112 and 0.033, all between-component). The four that do not: `g105` 1.18×10^{-2} and `g106` 3.85×10^{-2}, flat to +0.17% and +0.007% over the final 800 steps while their weight span grows to 10^{26} and 10^{28}; `g10` 5.2×10^{-2}; and one disconnected graph that falls 92% then holds at 1.39×10^{-3}. Transitivity manipulation (detail): on the second base graph the high-T arm falls 588× between steps 250 and 400 and is still falling; its window rates are -2.4×10^{-3} per step over 125–250 and -2.6×10^{-2} over 250–400. The remaining differences—exponential versus Euler update, regularised versus exact transport—we tested directly. Running the first-order Euler update $d_e \leftarrow (1 - \alpha\kappa_e)d_e$ of [23] on the seven networks with a 200-step exponential baseline, at two step sizes and with and without renormalisation (Supplementary Table S5), the two updates agree at the matched step size $\alpha = 0.05$: within-component residuals lie within a factor of three of the exponential baseline on all seven (median ratio 0.97, range 0.35–1.7), so the residual at our horizon is not produced by the update rule. At $\alpha = 0.5$ it collapses by six to seventeen orders of magnitude on five of seven and by a factor of eight on a sixth, to $\leq 1.0 \times 10^{-9}$; the seventh (`g105`) had reached 1.4×10^{-3} in 121 steps and was still falling when the budget stopped it, and the unrenormalised variant behaves the same, so the collapse is not accounted for by renormalisation in the tests we ran. This is a step-size result and we report it as one: at $\alpha = 0.5$ a single step changes an edge length by up to 50%, so the Euler scheme is no longer a small-step approximation to the same equation. Holding integrated flow time fixed instead of step count makes the point sharply—2,000 exponential steps at $\Delta t = 0.05$ is the same nominal time as 200 Euler steps at $\alpha = 0.5$, and at that horizon the exponential flow with exact transport has *not* collapsed (1.18×10^{-2} and 3.85×10^{-2} on the two ENZYMES graphs). Much of the 200-step residual is

transient in the sense that a coarser discretisation removes it, but it is not removed by waiting at $\Delta t = 0.05$, and we do not claim that it is.

(each run used a 200-step exponential baseline at $\Delta t = 0.05$; Supplementary Table S5.)

Detail for the exploratory sweeps and the sham control. Sham control (detail): over all 25 pairs the sham residual is higher than the real one in 24/25 at the matched horizon (one-sided sign test $p = 7.7 \times 10^{-7}$; Wilcoxon $p = 8.9 \times 10^{-8}$), with a median real residual of 0.127 against a median sham residual of 1.147; over the 21 pairs surviving a pre-specified exclusion the split is 21/21 ($p = 4.8 \times 10^{-7}$), and the single reversal in the all-pairs analysis is a case where the conclusion is unaffected either way. The permutation preserves the empirical curvature multiset exactly and destroys only the edge–curvature correspondence and its spatial arrangement.

Monotonicity of the variance (control). The update $w_e \leftarrow w_e e^{-\Delta t \kappa_e}$ does not imply $\mathrm{Var}[\kappa(t+1)] \leq \mathrm{Var}[\kappa(t)]$, so we report the increment statistics rather than assume them (Methods): across 6,052 increments, 10 of 16 trajectories never increase, the remaining 6 do so on 3.0–16.3% of steps with a largest single increase of 0.39%, and cumulative upward movement is at most 3.7% of the total decrease achieved. The variance is *empirically* near-monotone; we claim no monotonicity in general, and the stopping rule does not need one, since it fires on $|\Delta V|$, not on its sign.

Limitations of the design. Five limitations bound every statement above. *(i) Estimator.* V_{pooled} and V_{within} are not interchangeable on a disconnected graph; both are reported for every network, and the within-component re-run reached a median of only 86 steps, so those residuals are themselves finite-horizon quantities. *(ii) Transport.* The production curvature is a Sinkhorn surrogate at fixed $\varepsilon = 0.05$ and idleness $\alpha = 0.5$; the extreme weight ratios reached in Pattern II make the transport subproblem ill conditioned and put the surrogate out of range altogether (the kernel underflows), so the long-horizon residuals rest on exact transport alone, and we have tested but not eliminated a regularisation or precision origin for the residual. *(iii) Initial weights.* A positive third column read as a length is a modelling choice, not a verified data dictionary; the sensitivity control above bounds, but does not remove, its influence; for the eight datasets whose third column is a year or a constant the control is close to vacuous and is reported as such. *(iv) Independence.* The unit of independence is the source family, and inside the synthetic sweeps the two base graphs are shared by every condition, so the effect sizes there rest on 2 independent units and no more. *(v) Attribution.* The manipulations hold the degree sequence fixed but not motif structure, shortest paths or spectra, and the assortativity and community sweeps inherit the unsafe stopping rule, so their magnitudes carry horizon bias. Population C_4 covers 16 networks because a step costs 0.03–12 s and the corpus would need ~ 80 h to take all 93 to 400 steps: the sample is cost-stratified, not random, and biased towards small graphs. A lower bound on V_G^* in terms of a conflict between curvature-feasible intervals is conjectured, not proven.

Computational budget. All runs were executed on an 8-core x86-64 CPU with 9.9 GB of RAM, one thread per process (`OMP/OPENBLAS/MKL_NUM_THREADS=1`) and at most two concurrent processes. A single flow step costs a median of 3.3 s per network (range 0.02–57 s), so taking the full 93-network analysis set to 400 steps would cost ~ 81 h; the strict re-run is restricted to the 16 cheapest networks (C_4). Measured wall clock, each figure summed over the per-step timers written by the run itself: first pass (110 networks $\times$10 steps) 34 h; strict re-runs 2.0 h (1.5 h for the 16-network set reported here, 0.5 h for an auxiliary set with extended horizons); exact-transport runs 0.11 h; step-size sweep 0.08 h; ε-trajectory sweep 1.1 h. The within-component re-run of the 34 disconnected networks was not timed per run and is not reported. The exact-transport figure is small because it is not comparable with the corpus median: the nine networks of C_5 have

5–33 nodes and 6–69 edges, a step costs 5–21 ms there, and the 33,000 steps in question—nine networks taken to 2,000 steps, and three of the same networks continued to 5,000 steps—cost 387 s in total. Every horizon reported in this paper is a budget, not a choice, and is reported with the result.

Numerical reliability at extreme weight span. The 5,000-step exact-transport runs end with edge weights spanning 10^{65}–10^{72}, so we report what the residual at that span does and does not rest on (Supplementary Table S13). *Solvers.* Every transport problem is solved by network simplex (`ot.emd` from POT 0.9.7, `numItermax`= 10^6); the `scipy.linprog` HiGHS fallback exists only for the case in which POT raises, and it was never invoked—0 of 9,120 transport problems across the 60 checkpoints of the three extended runs fell back. Where we forced it, it agreed with the network simplex to a maximum relative cost difference of 1.2×10^{-15} while the weight span was below 10^5, degraded to 3×10^{-9}–6×10^{-8} at spans of 10^6–10^7, and to $O(1)$–$O(10)$ on a minority of edges beyond 10^8 (the *median* difference stays below 10^{-9} throughout). The fallback is not trustworthy at extreme span, and no reported number depends on it. *Conditioning.* Dividing each cost matrix by its own mean before solving—which multiplies the exact W_1 cost by the same factor and therefore leaves κ_e unchanged in exact arithmetic—reproduces every curvature to $\leq 2.2 \times 10^{-16}$ and the variance to a relative 3.6×10^{-16} at a span of 10^{26}–10^{29}. The magnitude of the costs, as opposed to their ratio structure, therefore costs nothing in double precision, and conditioning in double precision does not account for the residual. *Initial weighting.* Perturbing every initial length by 1% multiplicative jitter, five replicates per network and 1,000 steps of exact transport on `g105`, `g106` and `g10`, moves the terminal within-component residual by at most 2.4×10^{-5} in relative terms (Supplementary Table S13c), so sensitivity to the initial weighting does not account for the residual. *Consistency checks.* At every checkpoint the curvature of two reference graphs whose symmetry forces constant curvature (C_6 and K_6, unit weights) is recomputed and required to have zero variance: 60 of 60 checkpoints passed. (This check is applied to the reference graphs, not to the network under study, whose curvature is not constant at unit weights.) *The regularised solver.* At these spans the fixed-$\varepsilon = 0.05$ Sinkhorn kernel underflows and the surrogate is not usable: $\max_e |\kappa_e^{\text{sinkhorn}} - \kappa_e^{\text{exact}}|$ is of order 10^{16}–10^{25}. No Sinkhorn number is therefore reported beyond 2,000 steps, and the extended-horizon residual rests on exact transport alone. These controls bound three specific failure modes—solver agreement, conditioning in double precision, and sensitivity to the initial weighting. They do not establish that the residual is free of numerical error at these spans. The extreme weight ratio and the conditioning of the transport subproblems remain the central limitation; the constant-curvature check is applied to two reference graphs and not to the network under study; and the failure of two independent optimisers to find a lower-variance configuration is not a proof of global optimality. What we claim is that the tested numerical controls do not explain the residual.

What the extended-horizon runs do and do not establish. Three propositions are kept separate. *Observed:* for a fixed exact-transport algorithm, step size, idleness and initial weighting, the residual on three connected networks changes by less than 0.2% over 3,000 further steps of a 5,000-step run. *Not shown:* that the asymptotic limit is strictly positive—a trajectory that is flat over a finite window is consistent both with a positive limit and with a metastable transient longer than any horizon we could afford. *Not shown:* that the behaviour generalises to other networks of the same family, to other initial weightings, or to other step sizes. Supplementary Table S11 gives the per-network step count, residual, weight span and wall-clock time; the solver and the checkpoint diagnostics are those of the preceding paragraph, reported per checkpoint in Supplementary Table S13a.

Software. Python (NumPy, SciPy, NetworkX, pandas, matplotlib). All figures and statistics are produced by a single script (`make_paper_figures.py`) reading the raw result files; no number

in this paper is hard-coded.

Data availability

All 93 networks analysed here are publicly available from the Network Data Repository, http://networkrepository.com [44]. The repository distributes the data openly for research use and asks that it be acknowledged with the citation given in Ref. 44 and that derived or preprocessed versions be shared in turn (https://networkrepository.com/page_ack.php); reuse is otherwise subject to the terms stated on that site. Supplementary Table S1 gives, for every network of the analysis set, the dataset identifier, source family, size, number of components, the landing page `https://networkrepository.com/¡id¿.php` and the verified direct download URL, so that the corpus can be re-downloaded exactly; a fetch script is included in the code package. No new network data were generated. Derived data—the network manifest (initial curvature statistics, every numerical setting and the random seed), the per-dataset trajectories, the within-component decomposition of Supplementary Table S3 and all intermediate result files—are provided with this submission as Supplementary Data, together with Supplementary Tables S1–S17, and are archived alongside the code at https://doi.org/10.5281/zenodo.23330125.

Code availability

All analysis code is released under the MIT licence and accompanies this submission as a single archive (Supplementary Data), frozen at version `v1.0.1`. The archive contains the flow implementation, every experiment and analysis script, the machine-readable configuration (`config.yaml`, which records the unsafe legacy stopping rule as well, so that the artefact itself can be reproduced), the network manifest, the fixed random seeds, the raw result files, the supplementary tables and the figure-generation script, and the scripts that generate the stopping-rule and trajectory-audit tables (`S15`, `S17`). No number in the paper is hard-coded: every figure and statistic is regenerated from the raw result files by `make_paper_figures.py`, which also emits a log giving the number behind each plotted point. The release is available at https://github.com/fanqimcse/ricci-flow-finite-time-residuals (tag `v1.0.1`) and is archived under the concept DOI https://doi.org/10.5281/zenodo.23277161, which resolves to the current version [45]. We verified this rather than asserting it. The `v1.0.0` archive (https://doi.org/10.5281/zenodo.23277162) was first downloaded into an empty directory, and `make_paper_figures.py` was run there against the archived result files alone. All eight figures were regenerated and every line of the emitted data log—84 lines of statistics, one per plotted quantity—matched the working copy exactly. That check was informative rather than ceremonial: run this way, the release revealed path assumptions that a working tree cannot expose. The same clean-room procedure was then repeated on the corrected `v1.0.1` tree (tag `v1.0.1`, version DOI https://doi.org/10.5281/zenodo.23330125) that this submission freezes: copied into an empty directory and run against its archived result files alone, `make_paper_figures.py` again regenerated all eight figures and emitted a 93-line data log identical line for line to the working copy. We report these outcomes rather than a bare reproducibility claim. An archival request has also been submitted to Software Heritage (save request `2549855`, accepted), whose intrinsic identifier is issued once the archival visit completes.

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Author contributions

Qi Fan is the sole author and carried out all aspects of the work: conceptualisation, data curation, formal analysis, investigation, methodology, software, validation, visualisation, and writing (original draft and review).

Competing interests

The author declares no competing interests.

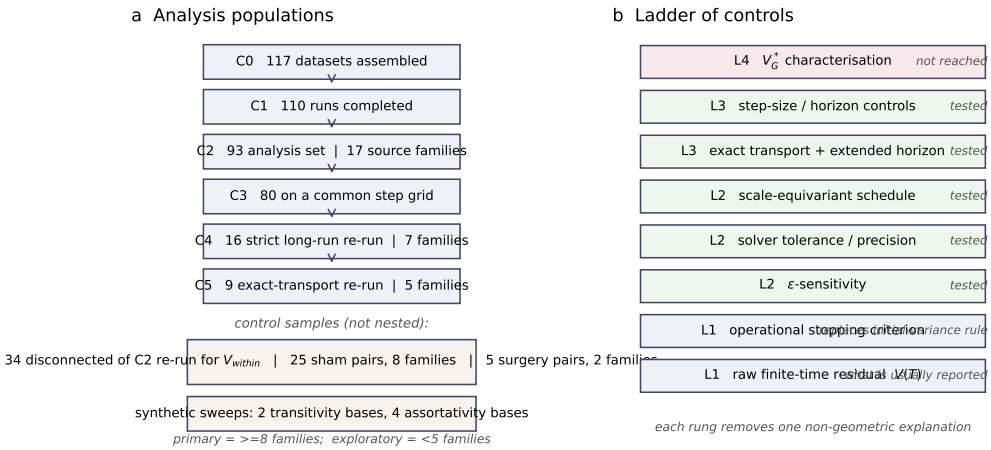


Figure 1: **Analysis populations and the ladder of controls.** **a**, Analysis populations. Nested subsets of the corpus, named C0–C5 and used consistently; each box gives the number of networks and, where inference is drawn, of independent source families. C4 is a cost-stratified convenience sample (all 16 below 0.76 s per step, cost ranks 3–22 of 113), not a random sample, and no prevalence claim is made from it. Of C2, 34 networks are disconnected and all were re-run under the within-component estimator (Supplementary Table S3); the 25 sham pairs and 5 surgery pairs are separate control samples drawn from the same corpus. **b**, The ladder of controls. Each rung removes one non-geometric explanation for an observed residual, from the raw finite-time value (L1) to the exact-transport, long-horizon and step-size controls that approach L3. The top rung, characterisation of V_G^* (L4), is not reached: no control in this paper bounds the geometric infimum.

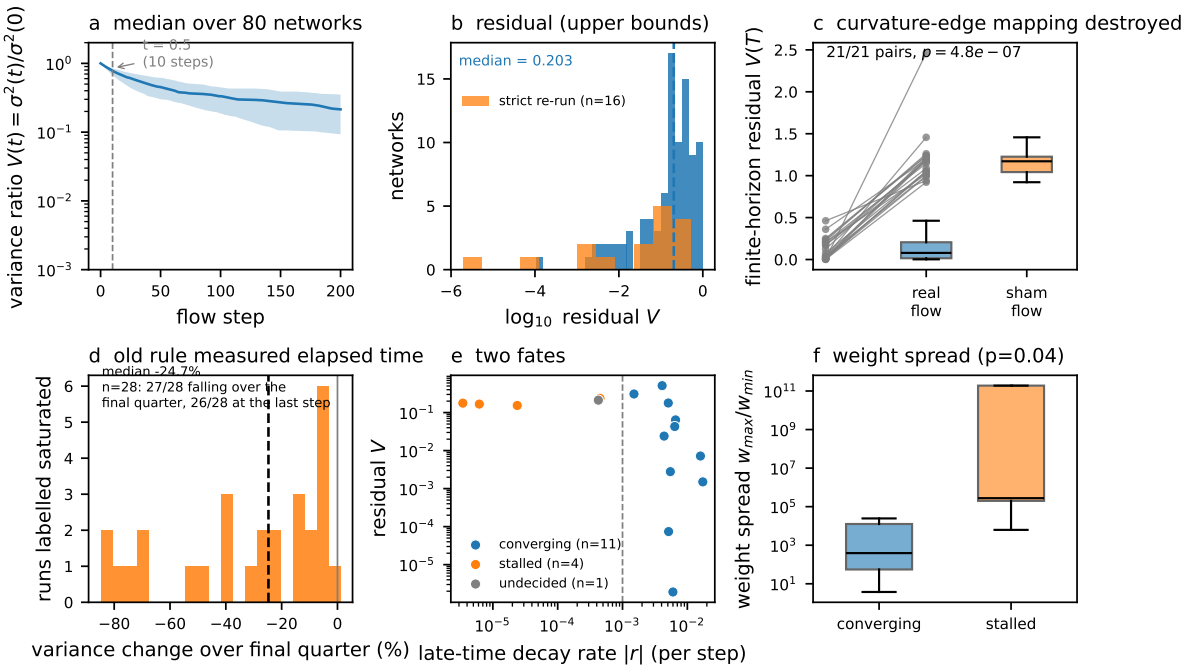


Figure 2: **The stopping rule decides the story, and the runs separate into two finite-horizon relaxation patterns. a**, Median (line) and interquartile range (band) of the variance ratio $V(t)$ over 80 networks with usable trajectories, on a common step grid; the 10-step transient used in earlier work sits at 0.78. **b**, Residual at the end of the run for the whole corpus ($n = 93$, terminal finite-horizon residuals) and for the 16 networks re-run to the longer horizon (orange); the corrected rule fired on one of them. **c**, Sham control: paired residuals with the curvature–edge mapping randomly permuted each step ($n = 25$ pairs, sham higher in 24/25, $p = 7.7 \times 10^{-7}$; 21/21 on the pre-specified subset). **d**, Variance change over the final quarter of the 28 runs the old rule labelled "saturated": 27 of 28 fell over the final quarter, 26 of 28 at the terminal step. **e**, Residual against the late-time decay rate for the strict re-run, separating Pattern I (blue) and Pattern II (orange); the dashed line marks a descriptive 10^{-3} per step rate, used only to make the two bands visible, and it is not a decision boundary. **f**, Edge-weight spread at the end of the run by pattern (Mann–Whitney $p = 0.04$ at the network level; family cluster-bootstrap interval for the median difference 2.85 decades, $[2.64, 5.25]$). Two of the four Pattern II networks are disconnected, and for both the residual is a between-component term (Table 1).

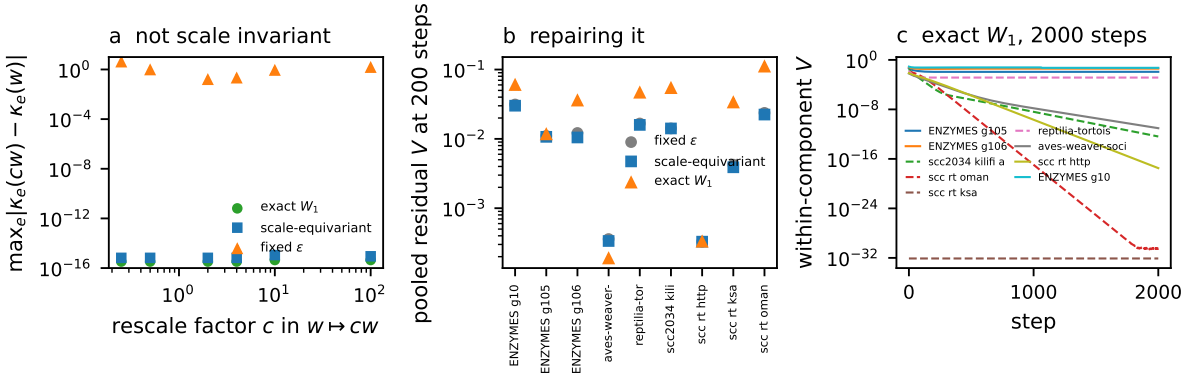


Figure 3: **The regularised curvature is not scale invariant, and repairing it does not remove the residual.** **a**, Maximum curvature change under a global rescale $w \mapsto cw$, over nine networks, two phases and six rescale factors. Exact W_1 transport and the scale-equivariant schedule are invariant to machine precision; the fixed-ε solver used throughout the paper is not, by up to 3.9 in a curvature whose range is $O(1)$. **b**, Residual at a matched horizon of 200 steps under the three estimators. **c**, Exact transport, no regularisation, 2,000 steps, within-component variance. Solid lines are connected graphs, dashed lines disconnected ones. Five of the nine fall below 10^{-11}—two connected (still falling at the last step) and three disconnected, whose pooled variance stays at 0.03–0.11. Four do not: three connected ENZYMES graphs hold 1.2×10^{-2} to 5.2×10^{-2}, two of them flat to $< 0.2\%$ over the final 800 steps while their weight span reaches 10^{26}–10^{28}, and one disconnected graph holds 1.4×10^{-3}. All three connected graphs were continued to 5,000 steps (Supplementary Table S11); the third, `g10`, flattened later—it was still falling by 14% between steps 1,000 and 1,500.

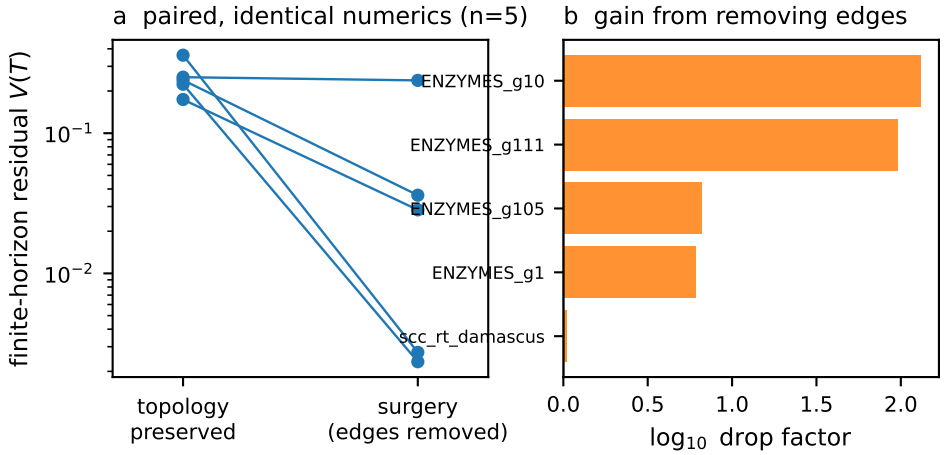


Figure 4: **Relaxing the edge-set constraint largely removes the residual.** **a**, Paired residuals with identical numerics; the constraint-relaxation arm removes only edges whose curvature stays above tolerance (4–14 edges of 41–84, bridges protected). Relaxed arm lower in 5/5 pairs ($p = 0.031$). **b**, Drop factor per pair (median 6.6×, maximum 132×). The intervention is outcome-informed—edges are selected by the curvature the flow produced—so this is a constraint-relaxation control, not a causal demonstration.

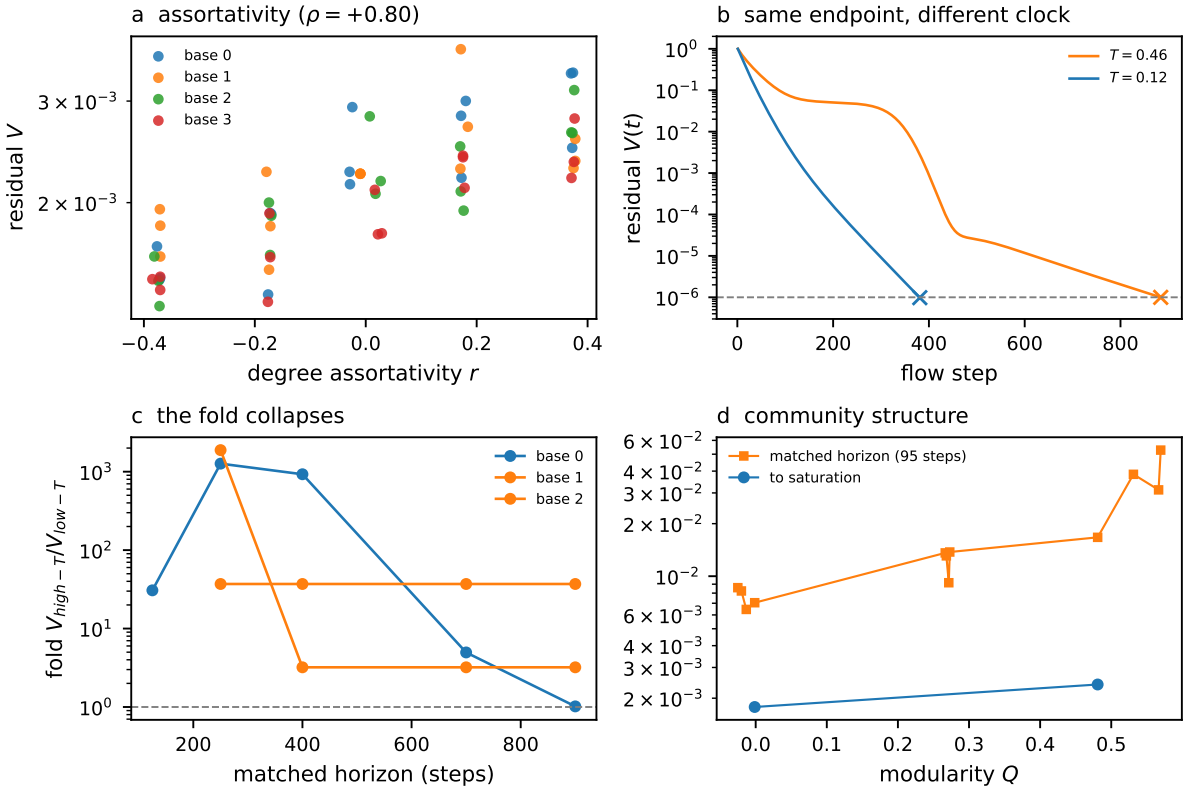


Figure 5: **Transitivity changes the clock, not the destination; community structure changes neither.** Degree sequences are held exactly fixed throughout. **a**, Residual versus manipulated assortativity ($n = 60$ saturated cells, 4 bases). **b**, Both arms of one base graph followed to a terminal state: raising transitivity from 0.12 to 0.46 leaves the endpoint unchanged—both arms reach the operational threshold $V = 10^{-6}$ (crosses at steps 381 and 884)—but multiplies the time to get there by 2.3; the dashed line is the floor. **c**, The fold between the two arms is a transient, not a plateau difference: 31× at 125 steps, 1.3×10^3 at 250, 9.3×10^2 at 400, 5.0 at 700 and 1.02 at 900 (base 0; bases 1 and 2 shown to their horizons). **d**, Community structure: strong at a matched 95-step horizon (6.1×) but only 1.35× at saturation—matched-horizon comparisons mistake slower convergence for a higher plateau.

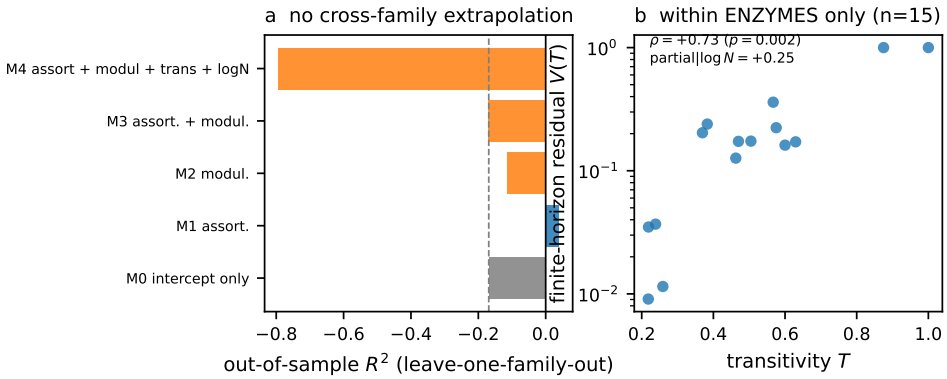


Figure 6: **The structure-to-residual mapping does not transfer across families. a**, Leave-one-family-out out-of-sample R^2; only assortativity rises marginally above the intercept baseline (dashed), and none of the models has usable predictive power. **b**, Within the largest family (ENZYMES, $n = 15$ at the operational threshold) the finite-horizon residual is associated with transitivity ($\rho = +0.73$) but the correlation collapses once size is controlled (partial $\rho = +0.25$, $p = 0.37$). **c**, Family-level cluster-bootstrap 95% intervals for the corpus correlations over 41 units and 11 families: the assortativity association survives clustering, the transitivity one does not, and the rate–residual interval includes zero.

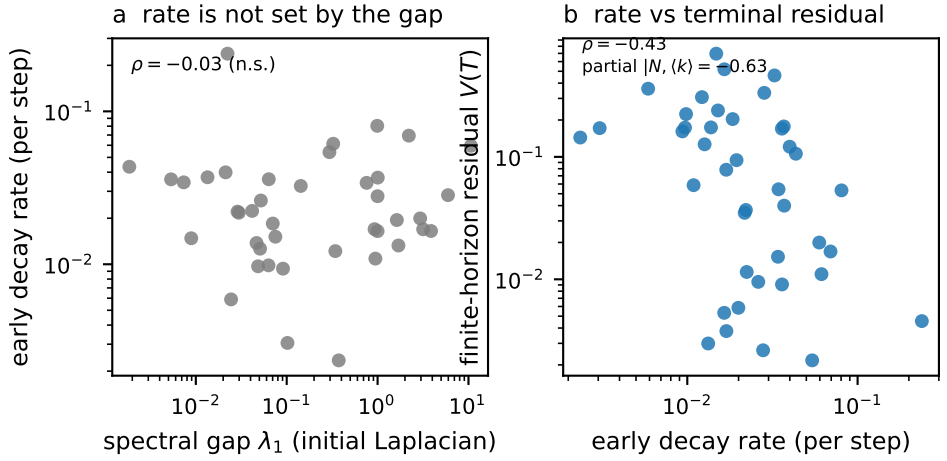


Figure 7: **Speed and depth are coupled; the spectral gap is not the clock. a**, Early decay rate versus initial Laplacian spectral gap across 41 units ($\rho = -0.03$, n.s.; cluster-bootstrap interval [-0.43,+0.39]); the gap does not set the rate. **b**, Observed decay rate is associated with the terminal finite-horizon residual at the network level ($\rho = -0.43$; partial $\rho = -0.63$, controlling size and mean degree); the family cluster-bootstrap interval [-0.68,+0.16] includes zero, so the association is not significant under family-level clustering.

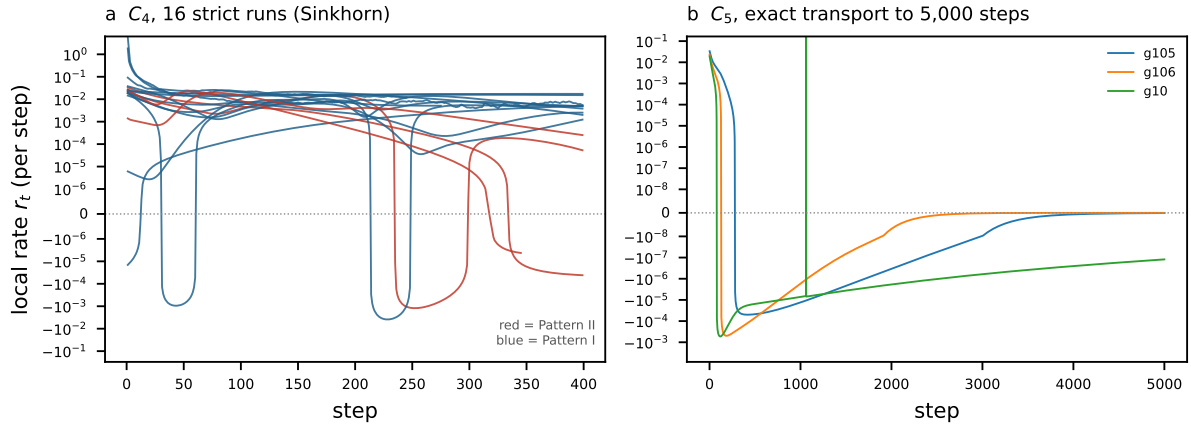


Figure 8: **The relaxation-rate trajectory, rather than a classification. a**, Local relaxation rate $r_t = -\log(V_t/V_{t-1})$ for the 16 runs of population C_4; $r_t > 0$ is decay, $r_t \approx 0$ a plateau, $r_t < 0$ a rebound (234 of 6,052 increments, 3.9%, are rebounds). Two bands are visible—one above 10^{-3} per step and one decaying towards 10^{-5}—but runs cross between them, so the bands are a description of the horizon, not of the limit. Red: Pattern II (terminal median rate below 10^{-4} per step); blue: Pattern I. **b**, Three connected ENZYMES networks under exact transport to 5,000 steps: $|r_t|$ falls below 10^{-7} per step and stays there while the weight span grows to 10^{65}–10^{72}. In all three r_t is *negative* at the end of the run: the variance is not decaying but drifting upwards, by at most +0.17% over the last 3,000 steps (Supplementary Table S11).